

Rainwater Harvesting Pilot: Final Report

Riverside Municipal Water Board

Report reference RMWB-2026-014, published 14 March 2026

1. Summary

Between 1 October 2025 and 31 December 2025, the Board ran a household rainwater harvesting pilot across four wards. 250 households participated. Average household water consumption fell by 18% against the same quarter in the previous year.

2. Background

Riverside has recorded four consecutive years of declining groundwater levels. The Board approved a limited pilot to test whether household-level harvesting units could reduce mains demand without changes to tariff structures.

Each participating household received one 2,000 litre collection unit, installed at no cost to the household. Total programme cost was Rs 42,00,000.

3. Results

Metric	Baseline	Pilot period	Change
Avg. household use (L/day)	412	338	-18.0%
Peak-hour demand (L/capita)	96.2	84.8	-11.4 L
Households reporting shortage	38%	21%	-17 pts
Units requiring repair	n/a	14 of 250	5.6%

Rainfall during the pilot period was 1,120 mm, which is 23% above the ten-year seasonal average of 910 mm.

4. Cost per household

Programme cost per participating household was Rs 16,800. At current mains water tariffs, the Board estimates a payback period of 6 to 8 years, though this estimate has not been independently audited.

5. Limitations

These results describe the pilot population only and do not automatically generalise to all households in Riverside.

Participation was voluntary. Households that volunteered may already have been more motivated to conserve water, so part of the observed reduction may reflect selection effects rather than the harvesting units themselves.

Rainfall during the pilot period was materially above average. A drier quarter would likely yield a smaller reduction.

No control group was used.

6. Recommendations

The Board recommends extending the pilot to 1,000 households across eight wards, including a matched control group, before any city-wide rollout is considered.

The Board further recommends that the repair rate of 5.6% be investigated before additional units are procured.

Contact: water.pilot@riverside-rmwb.example.gov.in